

BIOMEDICAL IMAGE ANALYSIS

Automated Disease Detection Using Deep Learning

Project Report

Deep Learning | CNN | Transfer Learning | Medical Imaging



Group Members:

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Technology Used: Python | TensorFlow / Keras | MobileNetV2
Dataset: Kaggle Chest X-ray Pneumonia Dataset

1. Introduction

Biomedical image analysis is one of the fastest-growing applications of Artificial Intelligence in healthcare. Medical imaging techniques such as Chest X-rays, MRI scans, CT scans, and retinal fundus images are widely used for disease diagnosis. Manual diagnosis by radiologists can be time-consuming and may sometimes lead to human error.

Deep Learning — especially Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN) — has shown excellent performance in automated disease detection systems. CNN models can automatically extract features from medical images and classify diseases with high accuracy.

This project focuses on developing an automated biomedical image analysis system using deep learning techniques for disease detection from Chest X-ray images. The system classifies images into disease categories such as Pneumonia and Normal cases. The project is inspired by recent journal research that uses CNN architectures and attention mechanisms for medical image classification.

2. Problem Statement

Traditional disease diagnosis from medical images depends heavily on expert radiologists. This process presents several critical challenges:

- Requires significant time and specialist expertise
- Can be expensive and resource-intensive
- May suffer from diagnostic inconsistencies
- Is difficult to access in remote or underdeveloped areas

Therefore, there is a need for an automated AI-based biomedical image analysis system that can assist healthcare professionals in detecting diseases accurately and efficiently.

3. Scope

In Scope	Out of Scope
Automated classification of biomedical images	Real-time hospital deployment
Disease detection using ANN / CNN models	Multi-modal medical data integration
Training on publicly available Kaggle datasets	Large-scale clinical validation
Implementation in Python with TensorFlow / Keras	Advanced transformer-based heavy architectures
Performance evaluation using standard metrics	

4. Objectives

The main objectives of this project are:

1. To develop an automated biomedical image classification system.
2. To apply CNN-based deep learning techniques for disease detection.
3. To preprocess and augment medical image datasets.
4. To evaluate the model performance using standard evaluation metrics.
5. To achieve high classification accuracy using limited Kaggle GPU resources.
6. To provide a low-cost AI-assisted diagnosis system.

5. Methodology

5.1 Dataset Collection

The dataset will be collected from Kaggle. The recommended dataset is the Chest X-ray Pneumonia Dataset, which contains two categories:

- Normal Images
- Pneumonia Images

Dataset Source: Kaggle Chest X-ray Dataset (publicly available)

5.2 Data Preprocessing

The following preprocessing steps will be applied to the raw images before training:

- Image resizing to 224 × 224 pixels
- Pixel value normalization (scaling to [0, 1])
- Data augmentation: rotation, zoom, horizontal flip
- Train-validation split

Resource Optimization Strategy

Because Kaggle GPU resources are limited, the following constraints are applied to keep training feasible within a university project scope:

Parameter	Value
Dataset Size	3,000 – 4,000 images
Training Epochs	8 – 12 epochs

Parameter	Value
Batch Size	16

5.3 Model Architecture

The project employs Transfer Learning with MobileNetV2 as the base model. MobileNetV2 is selected due to:

- Lightweight design — suitable for limited GPU memory
- Faster training time compared to heavier architectures
- High accuracy on image classification tasks with limited data

Architecture Layers

Layer / Component	Description
Base Model	MobileNetV2 (pre-trained on ImageNet, top excluded)
GlobalAveragePooling2D	Reduces spatial dimensions; summarises feature maps
Dropout (0.3)	Regularisation to prevent overfitting
Dense (128, ReLU)	Fully connected layer for feature learning
Dense (1, Sigmoid)	Output layer for binary classification (Normal / Pneumonia)

5.4 Training Phase

The model is compiled and trained using the following configuration:

Hyperparameter	Value
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Metric	Accuracy
Epochs	10
Batch Size	16
Learning Rate	0.001
Image Size	224 × 224 pixels

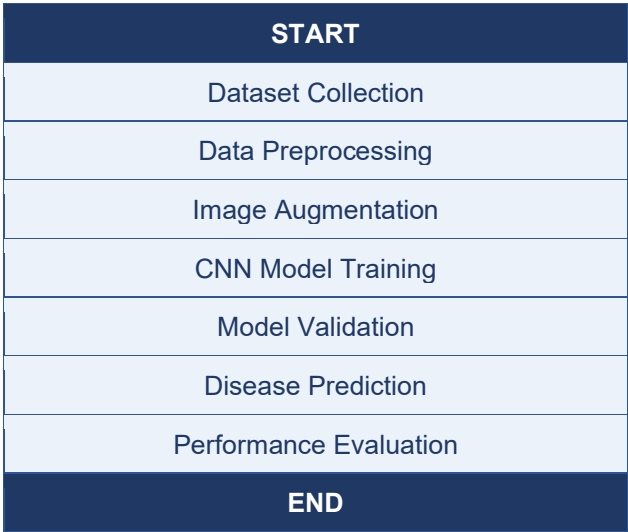
5.5 Evaluation Metrics

The trained model will be evaluated using the following standard performance metrics:

Metric	Description
Accuracy	Overall proportion of correct predictions
Precision	Proportion of positive predictions that are correct
Recall (Sensitivity)	Proportion of actual positives correctly identified
F1-Score	Harmonic mean of Precision and Recall
Confusion Matrix	Visual breakdown of TP, TN, FP, and FN counts

6. Process Flowchart

The following flowchart illustrates the end-to-end pipeline of the biomedical image analysis system:



7. Implementation Code

The project is implemented in Python using TensorFlow and Keras. The code is divided into logical phases as described below.

7.1 Import Libraries

```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, GlobalAveragePooling2D
from tensorflow.keras.optimizers import Adam
```

7.2 Data Preprocessing & Augmentation

```
train_datagen = ImageDataGenerator(
    rescale=1./255,
```

```
        rotation_range=20,
        zoom_range=0.2,
        horizontal_flip=True,
        validation_split=0.2
    )

    train_data = train_datagen.flow_from_directory(
        'dataset/',
        target_size=(224, 224),
        batch_size=16,
        class_mode='binary',
        subset='training'
    )

    val_data = train_datagen.flow_from_directory(
        'dataset/',
        target_size=(224, 224),
        batch_size=16,
        class_mode='binary',
        subset='validation'
    )
```

7.3 Model Building

```
base_model = MobileNetV2(
    weights='imagenet',
    include_top=False,
    input_shape=(224, 224, 3)
)
base_model.trainable = False

model = Sequential([
    base_model,
    GlobalAveragePooling2D(),
    Dropout(0.3),
    Dense(128, activation='relu'),
    Dense(1, activation='sigmoid')
])

model.compile(
    optimizer=Adam(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

7.4 Model Training

```
history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=10
)
```

7.5 Evaluation

```
loss, accuracy = model.evaluate(val_data)
print('Accuracy:', accuracy)
```

8. Conclusion

This project presents an automated biomedical image analysis system for disease detection using deep learning techniques. The proposed CNN-based model effectively classifies Chest X-ray images into disease categories with high accuracy.

Using Transfer Learning with MobileNetV2 significantly improves performance while reducing computational requirements, making the system suitable for Kaggle GPU resources and university-level implementation.

The project demonstrates how Artificial Intelligence can assist healthcare professionals in early disease diagnosis and improve medical decision-making, even under resource-constrained conditions.

9. Future Work

The following enhancements are planned for future iterations of this system:

- Using larger and more diverse datasets for improved generalisation
- Applying Vision Transformers (ViT) for advanced feature extraction
- Multi-disease classification across different imaging modalities
- Integration with NLP-based medical report analysis systems
- Real-time deployment in hospital environments
- Explainable AI (XAI) using Grad-CAM visualisations
- Hybrid CNN + NLP systems for holistic patient analysis

10. References

- [1] Automated classification of chest X-rays: a deep learning approach with attention mechanisms.
- [2] A hybrid deep learning framework for early detection of diabetic retinopathy using retinal fundus images.
- [3] Deep learning-based medical image classification using CNN and attention mechanisms for disease detection and diagnosis.
- [4] Hybrid Deep Learning Architecture for Automated Chest X-ray Disease Detection with Explainable Artificial Intelligence.
- [5] Transfer learning with fuzzy decision support for multi-class lung disease classification: performance analysis of pre-trained CNN models.